A Structural Threshold in Decision Capacity Governs Collapse in Self-Play Reinforcement Learning

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Abstract

We show that a threshold in decision capacity determines whether self-play reinforcement learning agents collapse under asymmetric rule perturbations. Across poker variants (Kuhn, Leduc, Leduc-4), matrix games (Matching Pennies), a dice game (Liar's Dice; 24,576 info sets), and four learning algorithms (Q-Learning, SARSA, REINFORCE, DQN), eliminating all *strategically meaningful* decisions causes rapid convergence to a deterministic exploitation attractor (DEA)—a fixed point at near-maximal loss. Preserving even a single contingent decision point prevents this collapse. A frozen baseline and fixed-opponent control confirm the mechanism is co-adaptation under constraint, not the perturbation itself. The phenomenon is timing-invariant, fully reversible upon action restoration, and intensifies under function approximation. These results establish a sharp, structural threshold between zero and minimal decision capacity that governs exploitability in self-play multi-agent systems.

Keywords

Multi-agent reinforcement learning; self-play; robustness; game theory; action-space perturbation

1 Introduction

Multi-agent reinforcement learning (MARL) agents trained through self-play have achieved superhuman performance in complex games [5, 14, 17], yet their robustness to structural changes in the environment remains poorly understood. Prior work has focused primarily on adversarial perturbations to observations or rewards [7], opponent modelling under distribution shift [6], or training stability in population-based methods [2]. Structural changes to the *action space*—where an agent permanently loses access to certain actions—remain largely unexplored.

Such perturbations arise naturally in practice. In robotics, hardware failures may disable actuators, eliminating actions mid-deployment. In financial trading, regulatory changes can restrict previously available strategies. In multi-agent software systems, API deprecations may remove action endpoints. Understanding how self-play agents respond to these asymmetric capability losses is essential for deploying RL systems where the action space is not guaranteed to remain static.

We study this question in imperfect-information games by removing one player's ability to bet or raise at subsets of decision nodes. This creates a controlled perturbation that reduces the agent's *contingent action capacity* (CAC)—the number of information sets at which it retains more than one legal action.

Our central finding is a sharp threshold effect. When CAC drops to zero (every decision is forced), adaptive self-play agents converge to a *deterministic exploitation attractor* (DEA)—a fixed point characterised by near-maximal loss and near-zero reward variance. When even a single decision point is preserved (CAC ≥ 1), agents stabilise near Nash equilibrium. A frozen baseline and fixed-opponent control isolate continued co-adaptation as the mechanism. Our contributions:

- We identify a sharp CAC threshold governing collapse in self-play RL and provide formal propositions characterising the zero-contingency fixed point.
- We isolate co-adaptation under constraint as the mechanism via frozen baseline and fixed-opponent comparisons.
- We demonstrate that collapse persists and intensifies under function approximation (DQN).
- We replicate across five games (1–24,576 info sets), four algorithms, multiple perturbation schedules, and show full reversibility. Boundary conditions (IPD, Liar's Dice) sharpen the threshold definition.

2 Related Work

Solving imperfect-information games. CFR [19] and variants have solved increasingly large poker games [3–5, 13]. These systems assume a fixed game structure; we study what happens when it changes asymmetrically.

Self-play RL.. From TD-Gammon [15] through AlphaZero [14] and AlphaStar [17], self-play has driven major advances. NFSP [8] and PSRO [12] address stability, but self-play can cycle or exhibit non-transitivity [2, 11]. We identify a distinct failure mode: co-adaptation-driven collapse under asymmetric action-space constraints.

MARL robustness. The literature encompasses adversarial policies [7], opponent-learning awareness [6], and distributional robustness [18]. We perturb the action space itself, revealing a qualitative threshold that continuous perturbations cannot produce.

Action masking and constrained RL.. Action masking is standard in game AI [9], and constrained MDPs formalise restrictions [1]. The dynamic consequences of mid-training removal under self-play have not been studied.

Exploitability. The standard game-theoretic metric [10, 16]. We connect our threshold to the best-response structure of reduced games via formal propositions.

3 Background

Kuhn Poker. Three cards ($J < Q < K$), two actions, ante 1. Nash P0: $-1/18$. 12 info sets. **Leduc Poker.** Six cards, three actions, two rounds. Nash P0: ≈ -0.087. 288 info sets. **Leduc-4.** Four ranks, three suits (12 cards). 504 info sets. Nash P0: ≈ -0.096. **Liar's Dice.** 1-die: 13 actions, 24,576 info sets. 2-dice: 25 actions, 200,000+ info sets. **Coordination Game.** Cooperative: 2 agents match a random target, 3 actions, 10 rounds. **Negotiation.** Ultimatum: P0 offers a split (11 actions), P1 accepts/rejects, 5 rounds. **Matching Pennies.** Single-shot, Nash: 50/50, value 0.

Table 1: Kuhn, zero contingency (20 seeds).

Agent	Pre	Post	p	d
CFR	−0.060	−0.221	<0.0001	−6.9
Q-Learning	−0.041	**−0.926**	<0.0001	−42.0

Agents. CFR (frozen Nash), Q-Learning (tabular ε-greedy), QL-Frozen (frozen at perturbation), DQN (2-layer MLP), SARSA, REINFORCE.

4 Methodology

Each experiment: 20 seeds, 20,000 episodes (50,000 for DQN). Perturbation at midpoint.

CAC. Number of info sets where the perturbed agent retains > 1 legal action. Unweighted by reach probability (see Limitations).

Self-play. Single agent plays both roles; Q-values indexed by player-specific info states. Verified equivalent to separate agents (Appendix B).

Statistics. Paired t-tests, bootstrap 95% CIs (10,000 resamples), Cohen's d.

5 Theory

Proposition 1 (Zero-contingency exploitation). *Let G' be the reduced game with* $\mathrm{CAC}(P_0) = 0$. *Then $v_0(G')$ equals the best-response value against P_0's forced policy:* $v_0(G') = -\max_{\pi_1} \sum_z u_1(z) \cdot \rho^{\sigma_0^f, \pi_1}(z)$.

Proof. P_0 has no choices; the game is a single-agent optimisation for P_1 with a unique pure BR. By the minimax theorem, $v_0(G')$ is determined uniquely. □

Proposition 2 (Residual contingency bound). *If $|A(h^*)| \geq 2$ for some $h^* \in \mathcal{I}_0$, then P_1 cannot adopt the pure exploitation strategy of Prop. 1, and $v_0(G') \geq v_0^{\mathrm{Nash}}(G) - \delta(h^*)$.*

Proof. P_0's choice at h^* forces P_1 to mix or condition, preventing convergence to the zero-contingency point. □

Corollary 1. *The* $\mathrm{CAC} = 1 \to 0$ *transition qualitatively changes the BR structure: from strategic interaction to trivially computable pure exploitation.*

6 Experiments

6.1 The Phenomenon

Under zero contingency (Table 1), Q-Learning converges to the DEA at −0.926 (normalized: 0.27; $d = -42.0$) within four episodes. Under residual contingency (CAC= 1), Q-Learning drops only to −0.065 ($d = -0.8$) and stabilises.

6.2 The Threshold

The 0→1 jump is $\Delta = 0.86$; the 1→2 jump is $\Delta = 0.03$—a sharp threshold, not a gradient (Table 2).

Table 2: CAC sweep (Kuhn).

CAC		CFR	QL	Norm.
0	Zero-cont.	−0.221	**−0.926**	0.27
1	Residual	−0.053	−0.065	0.48
2	Full	−0.053	−0.034	0.49

Table 3: Algorithm invariance (Kuhn, 5 seeds).

Algorithm	Type	Post	d
Q-Learning	Tab.	**−0.927**	−66.1
SARSA	Tab.	**−0.927**	−66.1
REINFORCE	Tab.	**−0.500**	−39.6
DQN	Neural	**−0.994**	−24.4

Table 4: Cross-game collapse severity.

Game	QL Post	Info sets	Norm.
Match. Pennies	−0.851	1	0.07
Kuhn	−0.926	12	0.27
Leduc	−0.252	288	0.49
Leduc-4	−0.185	504	0.49
Liar's Dice (1d)	−0.032*	24k	0.48
Coordination	+1.44‡	–	–
Liar's Dice (2d)	+0.008*	200k+	0.50

*Challenge-only retains strategic value. ‡Cooperative; degradation without collapse.

6.3 The Mechanism

Frozen baseline. QL-Frozen: −0.141 vs Q-Learning: −0.927 ($p = 0.004$, $d = -2.6$). Co-adaptation, not the constraint, drives collapse.

Fixed opponent. Against static Nash: −0.228 (no collapse). Self-play: −0.926. Co-adaptation is necessary.

6.4 Generalisation

All collapse (Table 3). DQN collapses hardest—policy entropy drops to near zero and Q-value gaps spike post-perturbation, confirming rapid deterministic convergence.

Collapse holds across poker and matrix games (Table 4). Liar's Dice shows no collapse—a boundary condition.

6.5 Boundary Conditions

IPD. Removing "cooperate" pushes toward Nash (always-defect). No collapse.

Liar's Dice. Challenge-only retains strategic value (contingent on private die). No collapse at either scale (1d/2d). Confirms the threshold depends on *strategic flexibility*, not action count.

Non-zero-sum domains. In the cooperative Coordination game, forcing P0 to a single action (CAC = 0) degrades performance ($p = 0.001$) but does not produce convergence to the DEA. In the Negotiation game, P1 shifts toward rejection but does not converge to unconditional rejection. Cooperative dynamics produce bounded degradation, not collapse—the threshold interacts with interaction structure.

Timing. Episodes 3k/10k/17k yield identical collapse ($-0.926/-0.927/-0.925$).

6.6 Dynamics

Recovery. Restoring actions produces full recovery ($\Delta = +0.90$) within four episodes—symmetric with collapse. The DEA is a maintained attractor, not corrupted representation.

Exploitability. Exact BR computation confirms post-perturbation exploitability spikes to maximum.

Variance. Post-collapse $V_{\text{total}} = 5\times10^{-6}$: the DEA is a deterministic fixed point.

7 Conclusion

We identified a sharp threshold in contingent action capacity governing collapse in self-play RL. The finding is game-general (1–200,000+ info sets), algorithm-invariant (including DQN), co-adaptation-driven, timing-invariant, and fully reversible. Boundary conditions (IPD, Liar's Dice) show the threshold depends on strategic flexibility, not action count. Entropy regularisation does not prevent collapse (Appendix C).

Limitations. CAC is unweighted by reach probability. Reversibility may not hold under deeper networks.

Future work. Larger games with neural CFR; cooperative settings; formal exploitability bounds; reach-weighted capacity.

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A Hyperparameter Sensitivity

	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.3$
$\varepsilon = 0.05$	−0.974	−0.976	−0.976
$\varepsilon = 0.15$	−0.924	−0.927	−0.927
$\varepsilon = 0.30$	−0.849	−0.853	−0.851

Table 5: Collapse across all hyperparameter combinations.

B Separate Self-Play Verification

Independent P0/P1 Q-tables: -0.926 ± 0.003 vs shared: -0.927 (diff: +0.0008).

C Entropy Regularisation

τ	QL Post
0.00	−0.927
0.05	−0.925
0.10	−0.923
0.20	−0.926

Table 6: Entropy regularisation has no effect on collapse.